Q1: 16 March (Shift 1) - Numerical

Let the curve $y=y(x)$ be the solution of the differential equation, $\frac{\mathrm{dy}}{\mathrm{dx}}=2(\mathrm{x}+1)$. If the numerical value of area bounded by the curve $\mathrm{y}=\mathrm{y}(\mathrm{x})$ and x -axis is $\frac{4\sqrt{8}}{3}$, then the value of $\mathrm{y}(1)$ is equal to___

Q2: 16 March (Shift 2) - Single Correct

Let C_1 be the curve obtained by the solution ofdifferential equation $2\mathrm{xy}\frac{\mathrm{dy}}{\mathrm{dx}}=\mathrm{y}^2-\mathrm{x}^2, \mathrm{x}>0$ & Let the curve C_2 be the solution of $\frac{2\mathrm{xy}}{\mathrm{x}^2-\mathrm{y}^2}=\frac{\mathrm{dy}}{\mathrm{dx}}$.

If both the curves pass through $(1,1)$, then the area enclosed by the curves C_1 and C_2 is equal to :

(1) $\pi-1$

(2) $\frac{\pi}{2}-1$

(3) $\pi+1$

(4) $\frac{\pi}{4}+1$

Q3: 17 March (Shift 2) - Numerical

Let $\mathrm{f}:[-3,1]\rightarrow \mathrm{R}$ be given as

$$f(x)=\begin{cases}\min\left\{(x+6),x^2\right\}, & -3\leq x\leq 0\\ \max\left\{\sqrt{x},x^2\right\}, & 0\leq x\leq 1\end{cases}$$

If the area bounded by $y=f(x)$ and x -axis is A, then the value of 6 A is equal to ____.

Q4: 18 March (Shift 2) - Single Correct

The area bounded by the curve $4y^2=x^2(4-x)(x-2)$ is equal to:

(1) $\frac{\pi}{8}$

(2) $\frac{3\pi}{8}$

(3) $\frac{3\pi}{2}$

(4) $\frac{\pi}{16}$

Q5: 18 March (Shift 2) - Numerical

Let $y=y(x)$ be the solution of the differential equation $x d y-y d x=\sqrt{\left(x^{2}-y^{2}\right)} d x, x \geq 1$, with $\mathrm{y}(1)=0$. If the area bounded by the line $\mathrm{x}=1, \mathrm{x}=\mathrm{e}^{\pi}, \mathrm{y}=0$ and $\mathrm{y}=\mathrm{y}(\mathrm{x})$ is $\alpha \mathrm{e}^{2 \pi}+\beta$, then the value of $10(\alpha+\beta)$ is equal to ___

Answer Key

Q1 (2) **Q2** (2) **Q3** (41) **Q4** (3)

Q5 (4)

Q1 (2)

$\frac{dy}{dx}=2(x+1)$

$\Rightarrow \quad \int dy=\int 2(x+1)dx$

$\Rightarrow y(x)=x^2+2x+C$

Area $=\frac{4\sqrt{8}}{3}$

$-1+\sqrt{1-C}$

$\Rightarrow \quad 2\int_{-1}^{-1+\sqrt{1-C}}\left(-(x+1)^2-C+1\right)dx=\frac{4\sqrt{8}}{3}$

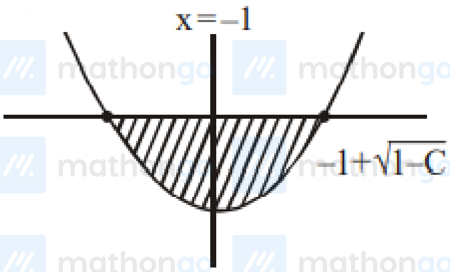
$\Rightarrow \quad 2\left[-\frac{(x+1)^3}{3}-Cx+x\right]_{-1}^{-1+\sqrt{1-C}}=\frac{4\sqrt{8}}{3}$

$\Rightarrow \quad -(\sqrt{1-C})^3+3c-3C\sqrt{1-C}$

$-3+3\sqrt{1-C}-3C+3=2\sqrt{8}$

$\Rightarrow \quad C=-1$

$\Rightarrow \quad f(x)=x^2+2x-1, \quad f(1)=2$



Q2 (2)

$\frac{dy}{dx}=\frac{y^2-x^2}{2xy}, \quad x\in(0,\infty)$

put $\quad y=vx$

$$x\frac{dv}{dx}+v=\frac{v^2-1}{2v}$$
$$\frac{2v}{v^2+1}dv=-\frac{dx}{x}$$

Integrate,

$$\ln\left(v^2+1\right)=-\ln x+C$$

$\ln\left(\frac{y^2}{x^2}+1\right)=-\ln x+C$

put $x=1, y=1, C=ln2$

$\ln\left(\frac{y^2}{x^2}+1\right)=-\ln x+\ln 2$

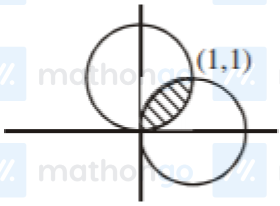
$\Rightarrow x^2+y^2-2x=0 \quad (\text{ Curve } C_1)$

Similarly,

$\frac{dy}{dx}=\frac{2xy}{x^2-y^2}$

Put $y=vx$

$x^2+y^2-2y=0$

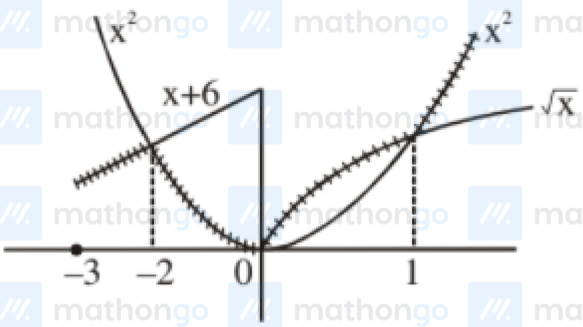


required area $=2\int_0^1\left(\sqrt{2x-x^2}-x\right)dx=\frac{\pi}{2}-1$

Q3 (41)

f : $[-3,1]\rightarrow$ R

$$f(x)=\begin{cases}\min\left\{(x+6),x^2\right\} & ,-3\leq x\leq 0\\ \max\left\{\sqrt{x},x^2\right\} & ,0\leq x\leq 1\end{cases}$$



area bounded by $y=f(x)$ and x -axis

$= \int_{-3}^{-2}(x+6)\mathrm{d}x + \int_{-2}^{0} x^2 dx + \int_0^1 \sqrt{x} dx$

$A = \frac{41}{6}$

$6 \mathrm{~A} = 41$

Q4 (3)

$4y^2 = x^2(4-x)(x-2)$

$|y| = \frac{|x|}{2}\sqrt{(4-x)(x-2)}$

$\Rightarrow y_1 = \frac{x}{2}\sqrt{(4-x)(x-2)}$

and $y_2 = \frac{-x}{2}\sqrt{(4-x)(x-2)}$

$\mathrm{D} : \mathrm{x} \in [2,4]$

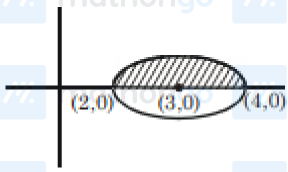
Required Area

$= \int_2^4 (\mathrm{y}_1 - \mathrm{y}_2)\,\mathrm{dx} = \int_2^4 \mathrm{x}\sqrt{(4-\mathrm{x})(\mathrm{x}-2)}\mathrm{dx}$ ...(1)

Applying $\int_{\mathrm{a}}^{\mathrm{b}} f(\mathrm{x})\mathrm{dx} = \int_{\mathrm{a}}^{\mathrm{b}} f(\mathrm{a}+\mathrm{b}-\mathrm{x})\mathrm{dx}$

Area $= \int_2^4 (6-x)\sqrt{(4-x)(x-2)}dx$ ...(2)

$(1)+(2)$



$2 \mathrm{~A} = 6\int_2^4 \sqrt{(4-\mathrm{x})(\mathrm{x}-2)}\mathrm{dx}$

$\mathrm{A} = 3\int_2^4 \sqrt{1-(\mathrm{x}-3)^2}\mathrm{dx}$

$\mathrm{A} = 3 \cdot \frac{\pi}{2} \cdot 1^2 = \frac{3\pi}{2}$

Q5 (4)

$xdy - ydx = \sqrt{x^2 - y^2}dx$

$\Rightarrow \frac{xdy - ydx}{x^2} = \frac{1}{x}\sqrt{1-\frac{y^2}{x^2}}dx$

$\Rightarrow \int \frac{d\left(\frac{y}{x}\right)}{\sqrt{1-\left(\frac{y}{x}\right)^2}} = \int \frac{dx}{x}$

$\Rightarrow \sin^{-1}\left(\frac{y}{x}\right) = \ell n|x| + c$

at $x = 1, y = 0 \Rightarrow c = 0$

$y = x\sin(\ell n x)$

$A = \int_1^{e^x} x\sin(\ell n x)dx$

$a = e^t, dx = e^t dt \Rightarrow \int_0^x e^{2t}\sin(t)dt = A$

$\Rightarrow$

$a = \frac{1}{5}, \beta = \frac{1}{5}$ so $10(\alpha + \beta) = 4$